Post-Quantum Cryptography Migration

Part 13 - Where It Runs: Enterprise Infrastructure & Storage

Compiled by the Spaced Research Ledger

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The machines that run and store enterprise data live longer than almost anything else in computing. A storage array, a tape cartridge, or a server management chip bought today will still be in service when quantum computers are far closer. That makes this hardware's cryptography a decision with a decade-long tail.

This report covers the layer in five sections. Section 1 covers the virtualization platforms, Section 2 the management controllers inside servers, and Section 3 backup and storage systems. Section 4 covers disk and data-at-rest encryption, and Section 5 the peripherals and badge readers at the physical edge.

The surprises run in both directions. Tape, the oldest medium in the room, shipped the world's first post-quantum-ready drive, building on demonstrations that predate the standards by five years. Server management moved hard, with HPE and Dell signing firmware with post-quantum algorithms and marketing quantum-resistant security processors. Even printers now ship post-quantum secure boot.

The laggard is the layer everyone assumes is fine. Data-at-rest encryption leans on the fact that symmetric AES is already quantum-resistant, but the key-wrapping and self-encrypting-drive standards around it show no post-quantum work at all. VMware, meanwhile, is honest that its timeline is hostage to the FIPS validation queue.

1. Virtualization Platforms

Hypervisors run most enterprise workloads, and their management traffic carries the credentials to everything. The platform vendors themselves have moved slowly, which has made backup vendors and third parties the practical delivery channel for post-quantum protection. This section covers a layer waiting on its suppliers.

Recent Developments

The motion comes from outside the hypervisors. wolfSSL announced a path to FIPS-validated post-quantum cryptography for Proxmox by replacing the platform's cryptographic backends wholesale [1].

Current State

Backup vendors became the common channel. Veeam's 13.1 release adds hybrid FIPS-plus-post-quantum key exchange for backup and replication traffic, keeping certified AES for bulk data, adoptable without redeployment [2]. Its hypervisor coverage spans essentially the whole market through a universal API [3]. Commvault moved even earlier, carrying the draft-era algorithms in an agility framework since 2024 and adding the backup KEM in 2025 [4].

VMware is candid about being stuck. Its own blog explains that post-quantum algorithms cannot ship in production until FIPS-certified builds of the underlying libraries exist, tying its timeline directly to the validation backlog [5]. The preparatory work is real: fleet-wide TLS profiles and a cryptographic bill of materials mapping every algorithm in use [5]. Nutanix shows no native feature, with a third-party encryption vendor filling the gap through certified node-level protection [6].

One dark footnote: a ransomware family brands itself with post-quantum terminology, actually using Kyber on Windows while its ESXi variant's quantum claim is simply false [7]. The attackers adopted the vocabulary before some vendors did.

2. BMC & Out-of-Band Management

Every server contains a second, hidden computer: the management controller that can reflash firmware and reach the console even when the machine is off. Compromise it and you own the fleet, which is why its firmware signing is racing ahead of nearly everything else. This section covers HPE, Dell, and the attestation standard beneath both.

Recent Developments

Firmware signing went post-quantum on both major lines. HPE's Gen12 servers sign the boot chain with LMS for the first step and ML-DSA for the rest, deliberately saying quantum-resistant rather than quantum-proof [8]. CNSA 2.0-approved signing is stated in product specifications [9]. Dell now dual-signs iDRAC firmware with a post-quantum signature beside the classical one, both validated on update [10].

The interfaces followed. HPE's iLO 7 Redfish service added ML-DSA-87 certificate types and a post-quantum enable flag [11][11]. The standards layer moved with them. The SPDM attestation protocol's working group opened feedback on hybrid support for version 1.5 [12] and published version 1.4.1 [13]. Its reference implementation carries the NIST algorithm set [14].

Current State

HPE claims the compliance high ground. Gen12 with iLO 7 reaches FIPS 140-3 Level 3, described as the first servers meeting NIST's quantum-resistant requirements [15], with a dedicated Secure Enclave processor guarding keys [15]. Further standards arrive over summer 2026 [16]. One marketing claim needs correction.

Framing larger RSA keys as being for post-quantum cryptography is technically wrong, since RSA of any size falls to a quantum computer [17]. That conflates classical hardening with quantum resistance [17].

Dell's approach is explicitly hybrid. iDRAC runs on FIPS-certified silicon with traditional and quantum-safe methods combined [18], and its newest generation adds a security enclave and component verification [19]. Attestation runs over SPDM with a silicon root of trust [20], and quantum-resistant firmware validation reaches production platforms in 2026 [21].

The attestation standard is the shared dependency. SPDM entered FIPS implementation guidance [22], its post-quantum expansion was announced by six standards bodies [23], and its full update waits on a standardized definition of hybrid encryption [24].

3. Backup & Storage Systems

Storage arrays and backup systems hold the data that harvest-now-decrypt-later attacks ultimately want. Their vendors split between shipping post-quantum protection on the wire and reframing existing AES as already quantum-safe. This section covers who did which, and the tape drive that beat everyone.

Current State

Tape got there first. LTO-10 is the first post-quantum-ready tape drive, carrying certificates for the NIST algorithms [25][26], with a higher-capacity cartridge keeping the same protection [27]. The lineage is real. IBM demonstrated Kyber and Dilithium on tape drives in 2019, five years before standardization, and commits its drives to the ratified algorithms [28].

It stands as one of only two products predating the 2024 announcement [29]. The key-management side is concrete, with drives connecting to IBM's key manager over post-quantum-protected TLS, upgraded in a specific order [30].

Dell shipped the flashiest first. Its PowerStore and PowerEdge combination with Broadcom's secure adapters delivers what it calls the first end-to-end in-flight post-quantum-ready encryption [31]. The specification is precise: ML-KEM-1024 key negotiation, ML-DSA-87 certificates, an LMS silicon root of trust, and line AES [32]. Its crypto module supports the finalized ML-DSA standard [33], and its own framing is sober, describing storage readiness as six layers rather than one upgrade [34].

NetApp embedded the algorithms in the array. It announced NIST-compliant post-quantum for file and block workloads [35], covering data at rest and in flight [36]. ONTAP supports all three algorithms for SSL, currently outside FIPS mode [37]. It also sells a separately named quantum-ready product that is AES-256 under the hood [38].

Partnerships extend the story, with F5 fronting its object storage with hybrid TLS [39][40] and a Taiwanese development agreement [41]. Pure Storage frames adoption as an extension of its continuous-upgrade model, a philosophy statement more than a shipped feature [38].

4. Disk & Data-at-Rest Encryption

Encrypted disks are the one place the quantum threat is smaller for real reasons: the bulk encryption is symmetric AES, which larger keys already protect. The soft spots are around the edges, in the key wrapping, the drive-access standards, and the databases. This section covers a layer where almost nothing has changed, and why that is only partly fine.

Recent Developments

The changes are at the perimeter. Microsoft named composite post-quantum support, ML-KEM included, as a future direction for BitLocker [42]. The Trusted Computing Group's new TPM guidance defines the quantum-ready hardware categories the drive-access chain will eventually rest on [43].

Current State

The standards under the drives are untouched. TCG Opal's authentication remains credential-based with no post-quantum algorithm mentioned in current documentation [44]. LUKS has no post-quantum update either, its most notable recent change being integration with Opal hardware that inherits whatever authentication is configured [45]. Where drive access is token-based, the exposure is inherited from the token's own readiness, still prototype-stage [45].

Databases show the same pattern with better marketing. Oracle's transparent data encryption defaults to AES-256, framed as strong against classical and quantum threats alike [46], with master keys in wallets and vaults integrating HSMs [47]. But no source confirms the key-wrapping mechanism itself has adopted a post-quantum KEM, as opposed to the database's network TLS, and the historical wallet mechanism is password-based [48]. For Microsoft SQL Server, nothing post-quantum-specific was found beyond classical key-manager integrations [49].

5. Peripherals & Physical Access

The migration's furthest edge is physical: the badge that opens the door, the printer in the hallway, the security key on a keyring. Some of it moved surprisingly fast, and some of it has not moved at all. This section covers printers with quantum-resistant chips, badges without any post-quantum at all, and the federal program planning the difference.

Recent Developments

Printers became an unlikely front-runner. Xerox's new A4 line ships hardware-enabled post-quantum secure boot on by default [50], with firmware updates extending protection to older devices back to 2018 [50]. The positioning against the RSA-and-ECC sunset is explicit [50]. Analysts rank it a print-security leader partly on that readiness [51].

HP answered with a quantum-resilient printer series built on quantum-resistant chips [52]. On the credential side, a partnership is building end-to-end post-quantum FIDO2 passkeys combining secure hardware with credential management [53].

The federal machinery started planning. GSA is expanding its evaluation lab to test quantum-resistant badges, passes, and access controls [54], and updating the federal identity architecture around crypto agility [55]. The phased timeline runs from discovery now to full migration by 2035 [56]. A hardware key now combines FIDO2 with badge credentials in one device [57].

Current State

The badge ecosystem is the untouched middle. HID's dominant credential platform documents symmetric encryption and mutual authentication with no post-quantum algorithm anywhere [58]. The company publishes awareness content rather than shipped features [59], and its identity-management platform's security story is workflow, not cryptography [60]. The design work exists as patents, a post-quantum Common Access Card concept [61] and a signature-free access scheme [62], neither a product.

Swissbit is the active corner, integrating HID credentials into its security key [63] and running a post-quantum evaluation platform for partners [63]. U.S. government guidance nonetheless lists identity hardware among categories with post-quantum-capable products widely available [64].

The printer race has a commercial logic. HP's commissioned study found no in-class competitor protecting firmware with quantum-resistant signatures at the time, a vendor claim [65]. Analysts locate the driver in procurement: printers bought today serve five to seven years, so quantum-resistant endpoints are entering RFPs, while attackers pivot toward printers as other endpoints harden [66]. Multi-generational refreshes at HP and Xerox's entry a year later suggest the category is now permanent [67].

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